

## Cars 2 Has an Alibi

Team Members: \_\_\_\_\_

Twenty-one Pixar films, each rated by three different services. Today you will accuse a movie of ruining the data. The evidence against it is real — but the test you run will not convict it, and working out why is the point.

Open `Class17_Cars_2_Has_An_Alibi_Skeleton.ipynb`. Answer here, on paper.

### Part 1. One Variable at a Time

18 minutes

Before comparing two variables you have to understand each one alone. The `stats()` method gives you six numbers per column.

#### 1.1 Fill in from the notebook:

|               | rotten<br>tomatoes | metacritic | critics<br>choice |
|---------------|--------------------|------------|-------------------|
| mean          |                    |            |                   |
| median        |                    |            |                   |
| mean – median |                    |            |                   |

1.2 One of those three columns has a mean far below its median; the other two barely differ. Which one, and what does a mean well below the median tell you about the shape of a distribution? (Sketch the shape if it helps.)

Now some vocabulary you need before the next plot. Order the values low to high and cut them into quarters.  $Q1$  is the median of the lower half,  $Q2$  is the median,  $Q3$  is the median of the upper half. The distance from  $Q1$  to  $Q3$  is the *interquartile range*, or IQR.

#### 1.3 By hand, no code. For these twelve numbers:

24, 25, 26, 27, 30, 32, 40, 44, 50, 52, 55, 57

$Q1 =$  \_\_\_\_\_  $Q2 =$  \_\_\_\_\_  $Q3 =$  \_\_\_\_\_  $IQR =$  \_\_\_\_\_

A *box plot* draws exactly that: a box from  $Q1$  to  $Q3$  with a line at the median. The whiskers reach the farthest point within  $1.5 \times IQR$  of the box, and anything past a whisker is drawn as a separate dot — a *flier*.

#### 1.4 Run `rmc.boxplot()`.

Which service gives the most favorable ratings overall? \_\_\_\_\_

The least favorable? \_\_\_\_\_

How many fliers appear in the whole plot? \_\_\_\_\_ On which service? \_\_\_\_\_

1.5 Find that film in the table and write down its name and its three ratings. Judging only by the box plot, would you call it an outlier? Say why.

## Part 2. Two Variables at a Time

18 minutes

2.1 Look at the scatter plot of `metacritic` against `critics_choice`. Is there a correlation? Positive or negative? Strong, moderate, or weak? Commit to an answer before you compute anything.

Standardizing turns any measurement into standard units — how many SDs from its own mean:

$$z = \frac{x - \bar{x}}{\sigma}$$

2.2 Plot the standardized ratings against each other. Compare the shape of the cloud to the unstandardized version. What changed, and what did not? Why?

The correlation coefficient  $r$  is the *average of the product* of the two variables in standard units. It has no units and always falls between  $-1$  and  $+1$ .

2.3 Think about the product  $z_x \times z_y$  for a single film before you compute anything. Fill in each cell with positive or negative:

|                     | $z_y$ above average | $z_y$ below average |
|---------------------|---------------------|---------------------|
| $z_x$ above average |                     |                     |
| $z_x$ below average |                     |                     |

Using that table, explain in one sentence why  $r$  comes out large and positive when the two variables track each other, and near zero when they do not.

**2.4 Compute  $r$  for all three pairs and rank them.**

rotten\_tomatoes & metacritic  $r =$  \_\_\_\_\_

rotten\_tomatoes & critics\_choice  $r =$  \_\_\_\_\_

metacritic & critics\_choice  $r =$  \_\_\_\_\_

**Which two services agree with each other most closely? Does that match what you guessed in 2.1?**

### **Part 3. The Alibi**

*14 minutes*

The film from Part 1 sticks out on the box plot and sticks out on the scatter plot. It is the obvious candidate for the point distorting your correlation. Put it on trial.

**3.1 For rotten\_tomatoes vs. metacritic:**

$r$  with all 21 films \_\_\_\_\_  $r$  with Cars 2 removed \_\_\_\_\_

**Removing the worst-looking point should have weakened the correlation. Which direction did it actually move?**

**3.2 The notebook removes each film one at a time and reports how much  $r$  changes.**

Largest effect: \_\_\_\_\_ change \_\_\_\_\_ Cars 2's change \_\_\_\_\_

**Find that film on the scatter plot and mark it. Describe where it sits compared to Cars 2.**

Cars 2 scored 57 on metacritic. The other low-rated films suggest a score near **46** would have been in keeping with the pattern. The notebook tries that hypothetical.

**3.3 Record all three:**

$r$  as observed \_\_\_\_\_  $r$  if Cars 2 had scored 46 \_\_\_\_\_  $r$  with Cars 2 removed \_\_\_\_\_

**Being out of line was worth about 0.06 of correlation. But removing Cars 2 gained only about 0.02. Roughly 0.04 went somewhere. Compare the spread of rotten\_tomatoes with and without Cars 2, and explain where it went.**

**3.4 So: is Cars 2 an unusual point? Is it the point that most affects  $r$ ? Explain how both answers can be what they are, in one or two sentences.**

Four small data sets, all with the same means, the same standard deviations, and the same correlation,  $r \approx 0.816$  — close to the correlation you found in 2.4.

**3.5 Plot all four. Which of them would you be comfortable summarizing with a correlation coefficient alone, and which would be actively misleading? Give a reason for each one you rule out.**